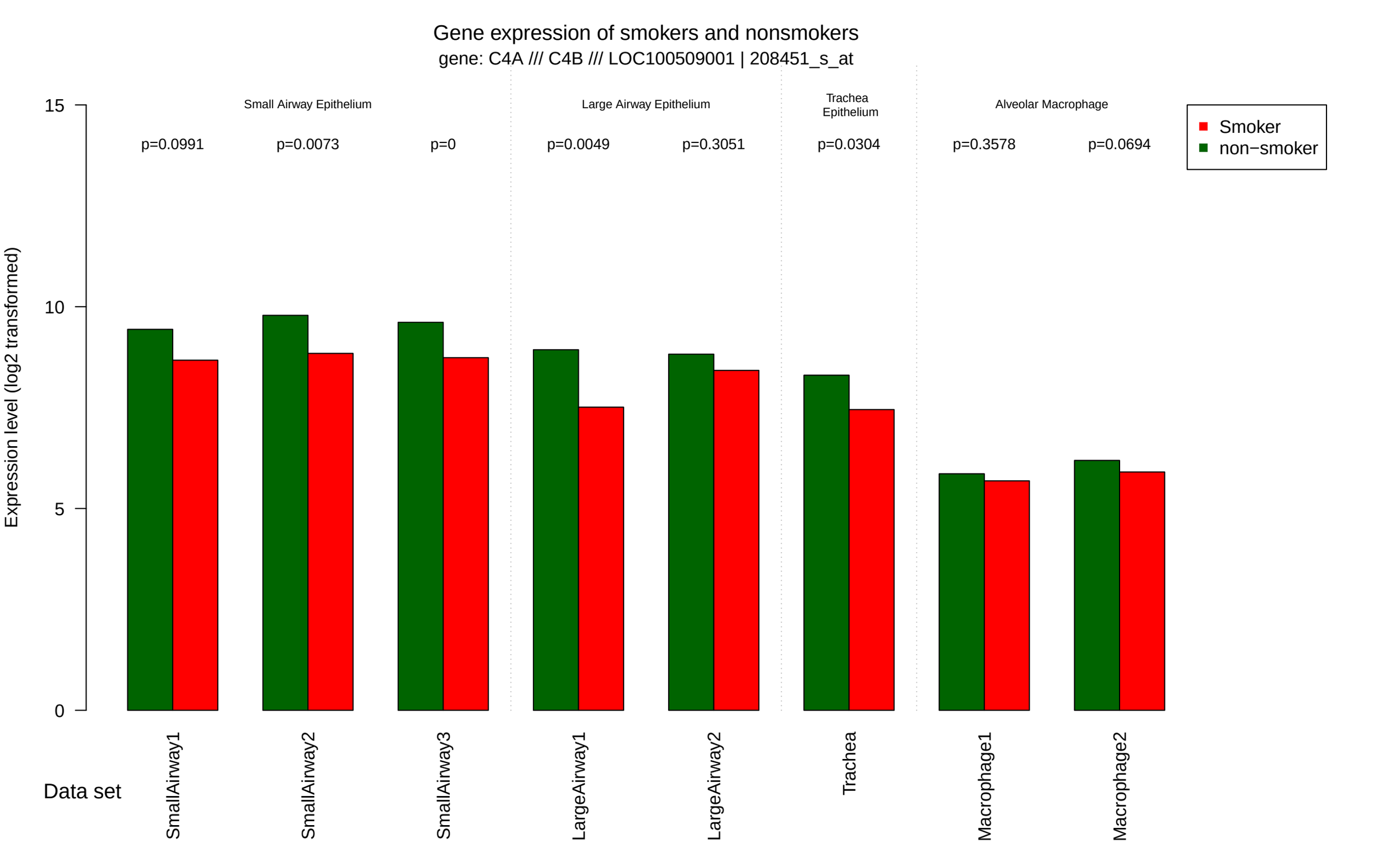
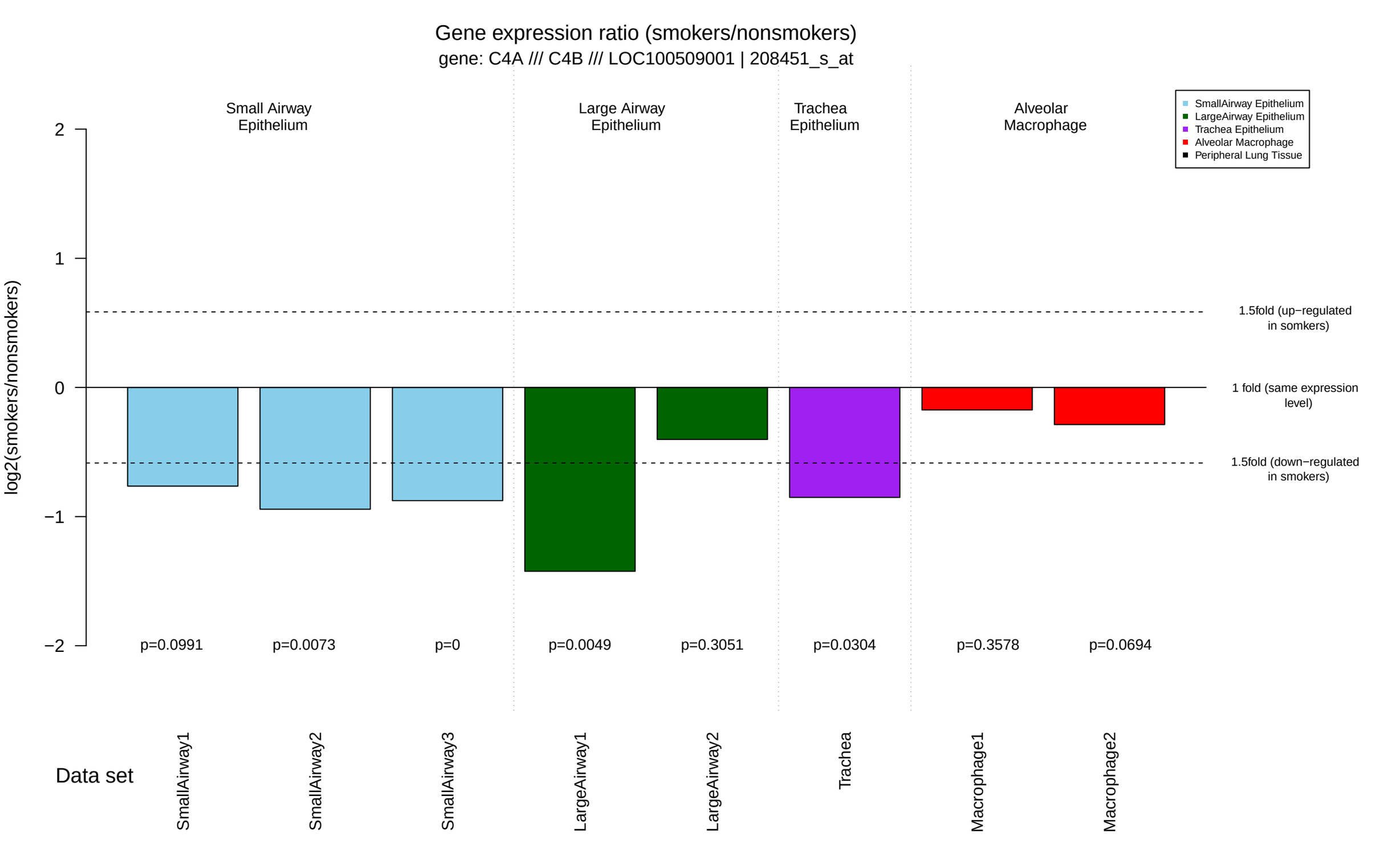
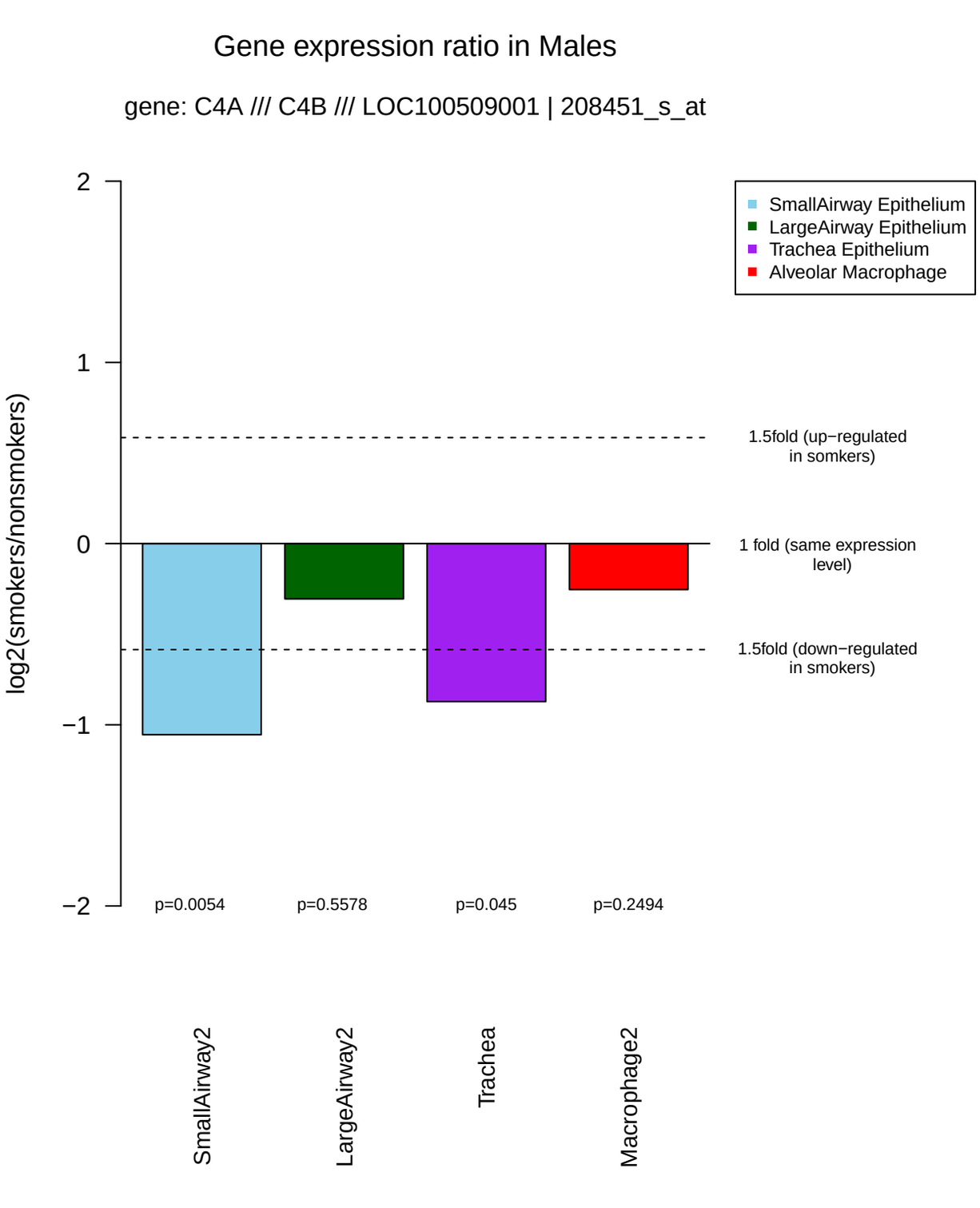
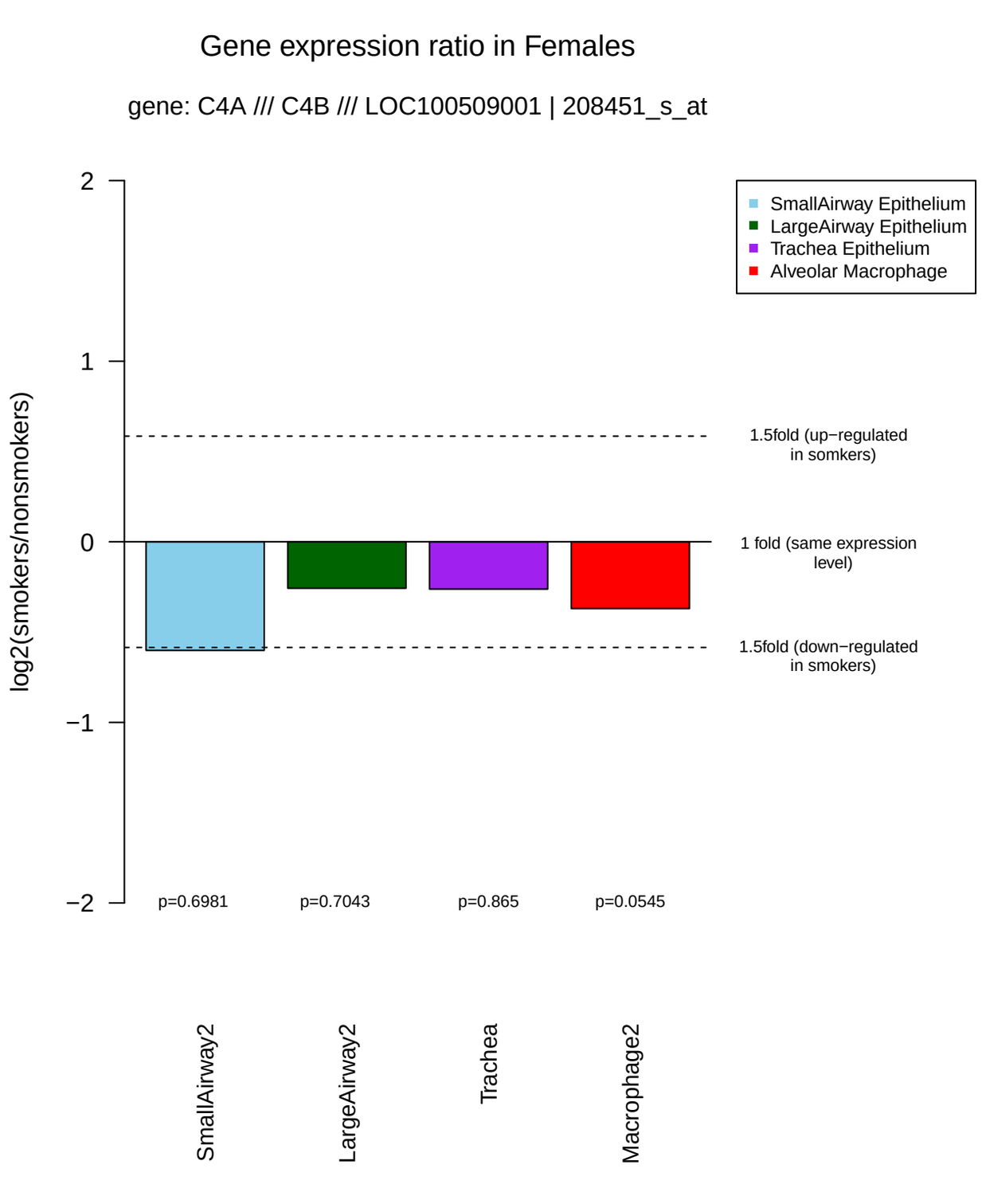
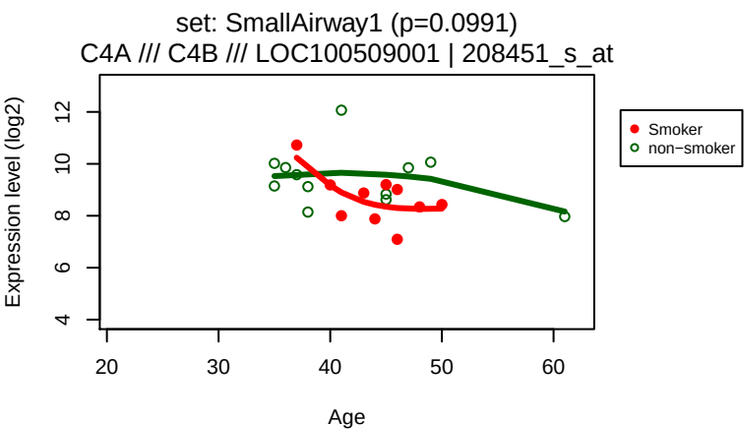
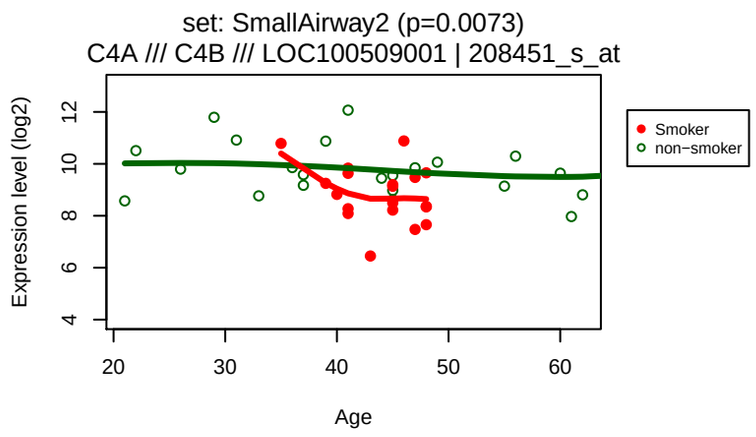

Gene expression of smokers and nonsmokers
gene: C4A /// C4B /// LOC100509001 | 208451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0991
p=0.0073
p=0
p=0.0049
p=0.3051
p=0.0304
p=0.3578
p=0.0694
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

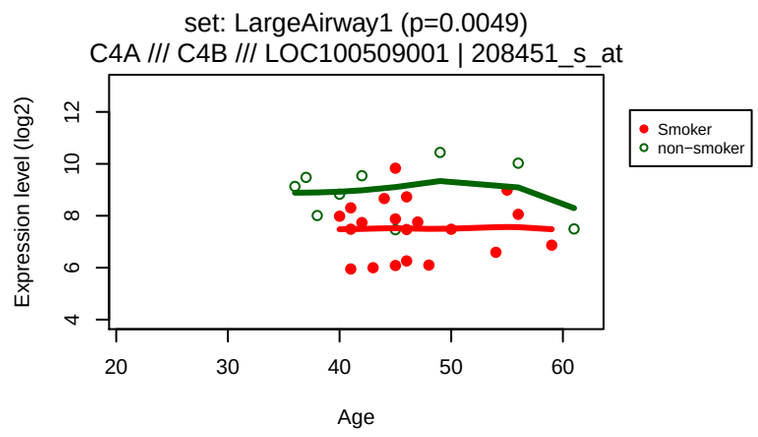

Gene expression ratio (smokers/nonsmokers)
gene: C4A /// C4B /// LOC100509001 | 208451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0991
p=0.0073
p=0
p=0.0049
p=0.3051
p=0.0304
p=0.3578
p=0.0694
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

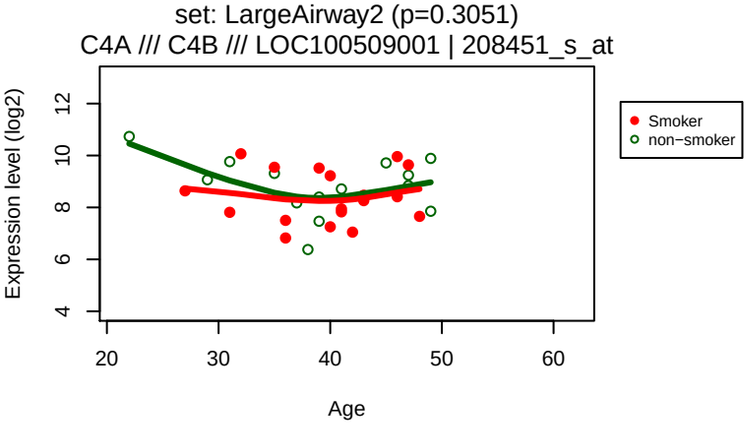

Gene expression ratio in Males
gene: C4A /// C4B /// LOC100509001 | 208451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0054
p=0.5578
p=0.045
p=0.2494
SmallAirway2
LargeAirway2
Trachea
Macrophage2

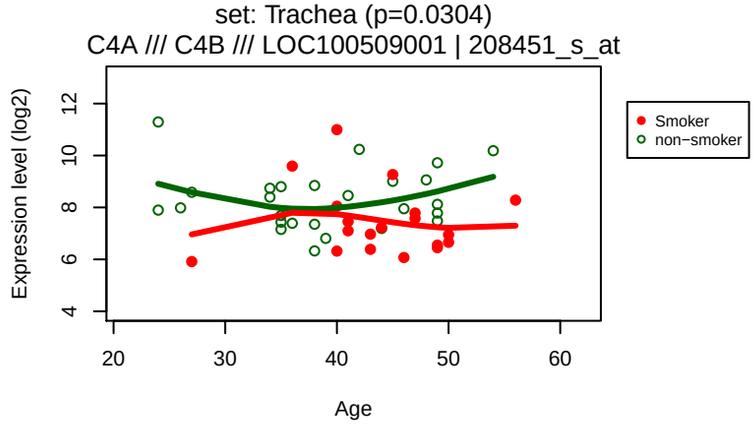

Gene expression ratio in Females
gene: C4A /// C4B /// LOC100509001 | 208451_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6981
p=0.7043
p=0.865
p=0.0545
SmallAirway2
LargeAirway2
Trachea
Macrophage2

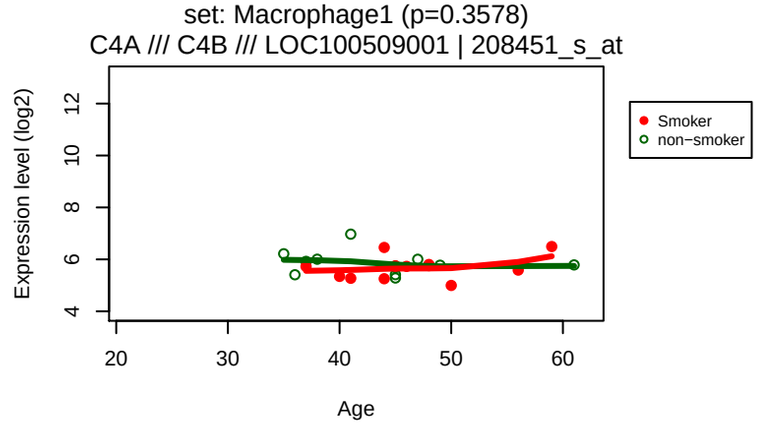

set: SmallAirway1 (p=0.0991)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker

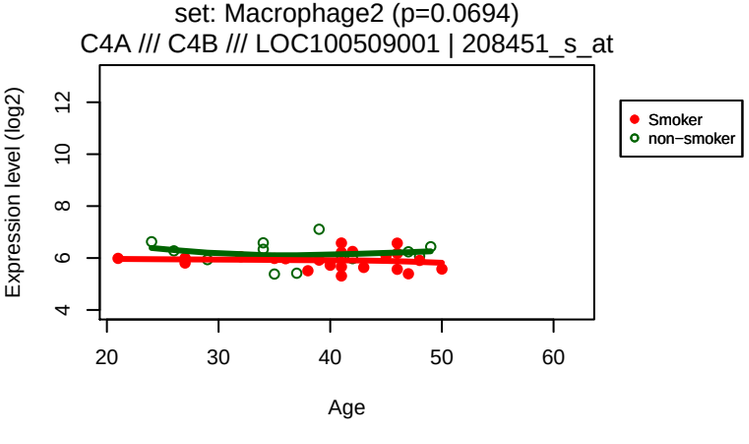

set: SmallAirway2 (p=0.0073)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker

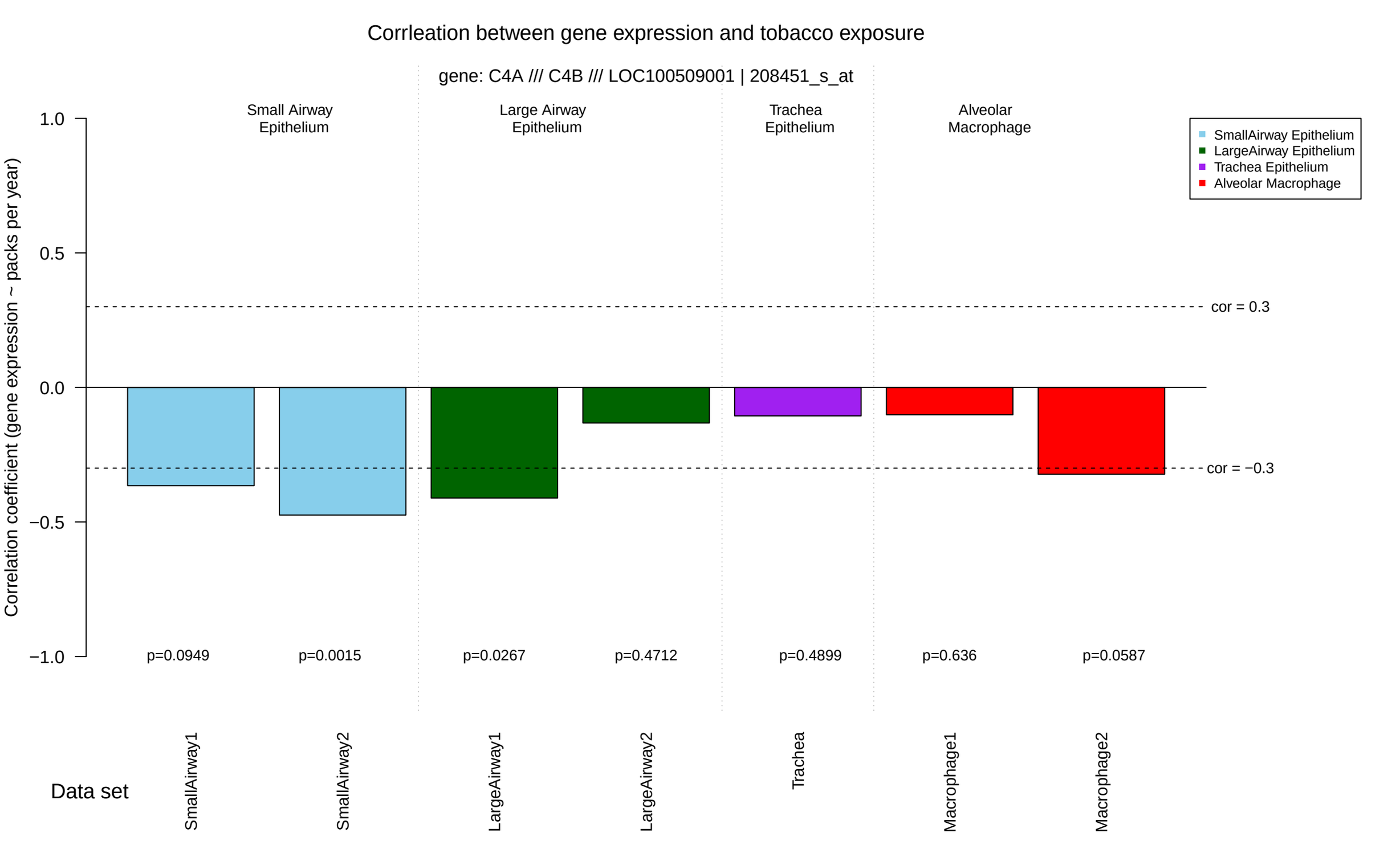

set: LargeAirway1 (p=0.0049)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.3051)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0304)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3578)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0694)
C4A /// C4B /// LOC100509001 | 208451_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: C4A /// C4B /// LOC100509001 | 208451_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0949
p=0.0015
p=0.0267
p=0.4712
p=0.4899
p=0.636
p=0.0587
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2